Lab 3: Architecture Justification

Corporate Mental Wellness & Counseling Platform **Narendra Kanchi | PES1UG24CS678**

Architecture Selection

We chose a Microservices Architecture for the Corporate Mental Wellness & Counseling Platform.

Architectural Choice

The platform is split into five small services, each owning its own database and matching one Lab 2 epic: **Anonymous Identity** (E1), **Booking** (E2), **Session** (E3), **Wellbeing Tracking** (E4) and **Wellness Analytics** (E5). An **API Gateway** is the single entry point for the Employee and Staff portals, and the external Notification and Telemedicine services (Lab 1 actors A-05 and A-04) sit behind interfaces. Layered and Client-Server were also considered. Both keep all data in one shared database and one deployable unit, which makes it harder to meet the privacy requirements and turns the server into a single point of failure.

Reason 1: Each feature area changes and depends on different things

The system relies on two external providers and was planned in Lab 2 as separate increments (Sprint 1: anonymous identity and booking; Sprint 2: sessions, notifications and check-ins). Because each service maps to one epic, it can be built, tested and deployed on its own. Replacing the video provider, for example, only changes the Session service behind its **ITelehealth** interface, and nothing else is redeployed.

Reason 2: A failure in one part must not stop employees getting help

Lab 1 (UC-03, alternate flow 8a) requires a booking to stay valid even when the notification service is down. With **INotify** as an asynchronous message queue, Booking keeps working and the confirmation is retried later. If the telemedicine provider fails, employees can still book and record check-ins. In a client-server design one server failure would block every function, including booking urgent support.

Security Advantage: data separation protects anonymity

The SRS requires that no employee can ever be linked to their counseling data (FR-001, FR-003, FR-006). Only the Anonymous Identity service handles pseudo-ID tokens, and every other service stores pseudo-IDs only, in its own database. A breach or a misused query in one service cannot join identities with session or wellbeing records, because no single database holds both. The gateway strips identifying headers (IP address, email) before forwarding requests, staff use a separate login-protected **IStaffAPI**, and Wellness Analytics receives only aggregates for groups of 10 or more through **IAggregates**, so HR cannot single out an employee (FR-004, FR-005).

Performance Benefit: independent scaling

Load is uneven. Video sessions are long-running and real-time, and bookings peak after working hours (FR-002 allows sessions during or after work). With microservices, extra instances of only the Session or Booking service are added at peak times instead of scaling the whole application. Sending notifications asynchronously also keeps the booking response fast, because the employee does not wait for an email or push message to be delivered.

Trade-off

Microservices add operational complexity: network calls between services, several databases and monitoring. This is kept manageable by limiting the design to five services with clear interfaces and a single gateway.